identifier over time. Although we have conducted robustness checks, in our preferred specification we do not include these job moves since we do not think that they map naturally to our model. For the dynamic model, we consider a subsample that covers the years 2001 to 2005. In addition to the criteria used to construct the 2002–2004 sample, we require that workers be fully employed in the same firm in 2001 and 2002, and in 2004 and 2005.

Descriptive Statistics

We now report descriptive statistics in the 2002–2004 and 2001–2005 samples, as well as in the subsamples of job movers. The numbers can be found in Table C1. The 2002–2004 sample contains about 600,000 workers and 44,000 firms. Hence, the average number of workers per firm is 13.7. The mean firm size as reported by the firm is higher, 37.6, due to our sample selection that focuses on fully employed male workers. In the 2001–2005 sample, the mean number of workers and mean reported size are 12.3 and 37.1, respectively. The distribution of firm size is skewed, and medians are smaller. At the same time, reported firm sizes in the subsamples of movers are substantially higher.

Identification relies on workers moving between firms over time. In the 2002–2004 sample, the mobility rate, which we define as the fraction of workers fully employed in 2002 and 2004 whose firm identifiers are different in these two years, is $19{,}557/599{,}775 = 3.3\%$. In the 2001–2005 sample, the rate is 2.4%. These numbers are lower than the ones calculated by Skans, Edin, and Holmlund (2009), who documented between-plant mobility

TABLE C1

DATA DESCRIPTION[a]

Years:	2002–2004 All	2002–2004 Movers	2001–2005 All	2001–2005 Movers
Number of Workers	599,775	19,557	442,757	9645
Number of Firms	43,826	7557	36,928	4248
Number of Firms ≥ 10	23,389	6231	20,557	3644
Number of Firms ≥ 50	4338	2563	3951	1757
Mean Firm Reported Size	37.59	132.33	39.67	184.77
Median Firm Reported Size	10	28	11	36
% High School Drop Out	20.6%	14%	21.5%	14.7%
% High School Graduates	56.7%	57.3%	57%	59%
% Some College	22.7%	28.7%	21.4%	26.3%
% Workers Younger Than 30	16.8%	28%	13.9%	23.8%
% Workers Between 31 and 50	57.2%	59%	59.4%	62.1%
% Workers Older Than 51	26%	13%	26.7%	14.2%
% Workers in Manufacturing	45.4%	35.1%	48.5%	40.4%
% Workers in Services	25.3%	33.7%	22.4%	27.8%
% Workers in Retail and Trade	16.7%	20.3%	16.3%	20.8%
% Workers in Construction	12.6%	10.9%	12.8%	11%
Mean Log-Earnings	10.18	10.17	10.19	10.17
Variance of Log-Earnings	0.124	0.166	0.113	0.148
Between-Firm Variance of Log-Earnings	0.0475	0.1026	0.0441	0.0947

[a]Descriptive statistics for a sample from the Swedish registry data that includes males, fully employed in the same firm in 2002 and 2004 (columns 1 and 2), and fully employed in the same firm in 2001–2002 and 2004–2005 (columns 3 and 4), for firms that are continuously present in the sample. The numbers in the table correspond to 2002.

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rates ranging between 4% and 6% between 1986 and 2000.[15] To understand how our sample selection influences the mobility rate, we have computed similar descriptive statistics in the entire 2002–2004 sample, without imposing that workers are fully employed in 2002 and 2004 or that firms exist in the two periods. Removing the requirements of full-year employment in both 2002 and 2004 and continuously existing firms results in a considerably less restrictive definition of mobility, as the mobility rate is 11.2% in this case. Although we prefer to focus on a more restrictive definition for estimation, as robustness checks we have also estimated the models on this larger sample, finding comparable results.

The between-firm log-earnings variance represents 38.3% of total log-earnings variance in 2002. This number is higher than the 31% percentage explained between plants in 2000, as reported by Skans, Edin, and Holmlund (2009). However, despite growing steadily over the past decades, the between-firm (or plant) component is still lower compared with other economies such as Germany, Brazil, or the United States. In Germany and Brazil, between-firm components are closer to 50%; see Baumgarten, Felbermayry, and Lehwald (2014) or Akerman, Helpman, Itskhoki, Muendler, and Redding (2013), for example. In the United States, Barth, Bryson, Davis, and Freeman (2014) reported a between-establishment log-earnings component of 46% to 49% in 1996–2007.

While differences across countries need to be interpreted cautiously due to differences in earnings definition and data collection, lower levels of between-firm earnings dispersion in Sweden are often attributed to historically highly unionized labor market and the presence of collective wage bargaining agreements. In particular, after World War II, the introduction of the so-called solidarity wage policy, which had as guiding principle "equal pay for equal work," significantly limited the capacity of firms to differentially pay their employees. However, several reforms over the last two decades have contributed to an increase in between-firm wage variation due to a more informal coordination in wage setting (see Skans, Edin, and Holmlund (2009), and Akerman, Helpman, Itskhoki, Muendler, and Redding (2013)). It is important to keep these features of the Swedish labor market in mind when interpreting our results.

Finally, comparing the first two columns—or the last two columns—of Table C1, we see that job movers are, on average, younger and more educated than workers who remain in the same firm. They also tend to work more in service sectors as opposed to manufacturing. At the same time, characteristics of job movers and stayers show substantial overlap.

REFERENCES

ABOWD, J. M., R. H. CREECY, AND F. KRAMARZ (2002): "Computing Person and Firm Effects Using Linked Longitudinal Employer-Employee Data," Unpublished Manuscript. [709]

ABOWD, J. M., F. KRAMARZ, P. LENGERMANN, AND S. PÉREZ-DUARTE (2004): "Are Good Workers Employed by Good Firms? A Test of a Simple Assortative Matching Model for France and the United States," Unpublished Manuscript. [701]

ABOWD, J. M., F. KRAMARZ, AND D. N. MARGOLIS (1999): "High Wage Workers and High Wage Firms," *Econometrica*, 67, 251–333. [699,718,722]

ABOWD, J. M., K. L. MCKINNEY, AND I. M. SCHMUTTE (2018): "Modeling Endogenous Mobility in Earnings Determination," *Journal of Business & Economic Statistics*, 1–14. [702,725]

[15]See their Figure 7.14. Skans, Edin, and Holmlund (2009) reported the fraction of workers employed in plants with more than 25 employees in years $t-1$ and t who changed plant between $t-1$ and t. As a comparison, in Germany, Fitzenberger and Garloff (2007) reported yearly between-employers transition rates of 7.5% in the period 1976 to 1996 for male workers.

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AIROLDI, E. M., D. M. BLEI, S. E. FIENBERG, AND E. P. XING (2008): "Mixed Membership Stochastic Blockmodels," *J. Mach. Learn. Res.*, 9, 1981–2014. [702]

AKERMAN, A., E. HELPMAN, O. ITSKHOKI, M.-A. MUENDLER, AND S. REDDING (2013): "Sources of Wage Inequality," *American Economic Review, Papers and Proceedings*, 103, 214–219. [736]

ANDREWS, M. J., L. GILL, T. SCHANK, AND R. UPWARD (2008): "High Wage Workers and Low Wage Firms: Negative Assortative Matching or Limited Mobility Bias?" *J. R. Stat. Soc. Ser. A Stat. Soc.*, 171, 673–697. [701, 722]

——— (2012): "High Wage Workers Match With High Wage Firms: Clear Evidence of the Effects of Limited Mobility Bias," *Econ. Lett.*, 117, 824–827. [701]

BAGGER, J., AND R. LENTZ (2014): "An Empirical Model of Wage Dispersion With Sorting," University of Wisconsin, Report. [700,705,712,722,726]

BARTH, E., A. BRYSON, J. C. DAVIS, AND R. FREEMAN (2014): "It's Where You Work: Increases in Earnings Dispersion Across Establishments and Individuals in the U.S.," Unpublished Manuscript. [736]

BARTOLUCCI, C., F. DEVICIENTI, AND I. MONZON (2015): "Identifying Sorting in Practice," Unpublished Manuscript. [712]

BAUMGARTEN, D., G. FELBERMAYRY, AND S. LEHWALD (2014): "Drivers of Wage Inequality in Germany: Trade, Technology, or Institutions?" Unpublished Manuscript. [736]

BECKER, G. S. (1973): "A Theory of Marriage: Part I," *Journal of Political Economy*, 81, 813–846. [700,701, 720]

BLEI, D. M., A. Y. NG, AND M. I. JORDAN (2003): "Latent Dirichlet Allocation," *J. Mach. Learn. Res.*, 3, 993–1022. [702]

BONHOMME, S. (2017): "Econometric Analysis of Bipartite Networks," in *The Econometric Analysis of Network Data*, ed. by B. Graham and A. De Paula. [702]

BONHOMME, S., AND E. MANRESA (2015): "Grouped Patterns of Heterogeneity in Panel Data," *Econometrica*, 83, 1147–1184. [702,712,732]

BONHOMME, S., K. JOCHMANS, AND J. M. ROBIN (2014): "Nonparametric Estimation of Finite Mixtures From Repeated Measurements," *Journal of the Royal Statistical Society, Series B* (forthcoming). [702]

BONHOMME, S., T. LAMADON, AND E. MANRESA (2017): "Discretizing Unobserved Heterogeneity," Unpublished Manuscript. [701,728]

——— (2019): "Supplement to 'A Distributional Framework for Matched Employer Employee Data'," *Econometrica Supplemental Material*, 87, https://doi.org/10.3982/ECTA15722. [702]

BOROVICKOVA, K., AND R. SHIMER (2017): "High Wage Workers Work for High Wage Firms," Unpublished Manuscript. [728]

BURDETT, K., AND M. G. COLES (2003): "Wage/Tenure Contracts and Equilibrium Search," *Econometrica*, 71, 1377–1404. [705]

BURDETT, K., AND D. T. MORTENSEN (1998): "Wage Differentials, Employer Size, and Unemployment," *Int. Econ. Rev.*, 257–273. [705]

CARD, D., J. HEINING, AND P. KLINE (2013): "Workplace Heterogeneity and the Rise of West German Wage Inequality," *Q. J. Econ.*, 128, 967–1015. [699,707]

DALY, M., D. HRYSHKO, AND I. MANOVSKII (2016): "Improving the Measurement of Earnings Dynamics," Unpublished Manuscript. [724]

DELACROIX, A., AND S. SHI (2006): "Directed Search on the Job and the Wage Ladder," *Int. Econ. Rev.*, 47, 651–699. [705]

DEMPSTER, A. P., N. M. LAIRD, AND D. B. RUBIN (1977): "Maximum Likelihood From Incomplete Data via the EM Algorithm," *J. R. Stat. Soc. Series B Stat. Methodol.*, 39, 1–38. [713]

EECKHOUT, J., AND P. KIRCHER (2011): "Identifying Sorting in Theory," *Rev. Econ. Stud.*, 78, 872–906. [700, 712]

FINKELSTEIN, A., M. GENTZKOW, AND H. WILLIAMS (2016): "Sources of Geographic Variation in Health Care: Evidence From Patient Migration," *Quarterly Journal of Economics*, 131, 1681–1726. [700]

FITZENBERGER, B., AND A. GARLOFF (2007): "Descriptive Evidence on Labor Market Transitions and the Wage Structure in Germany," *Int. Econ. Rev.*, 227, 115–152. [736]

FRIEDRICH, B., L. LAUN, C. MEGHIR, AND L. PISTAFERRI (2014): "Earnings Dynamics and Firm-Level Shocks," Unpublished Manuscript. [715,734]

GOLDSCHMIDT, D., AND J. SCHMIEDER (2015): "The Rise of Domestic Outsourcing and the Evolution of the German Wage Structure," *Quarterly Journal of Economics*. To appear. [699,700]

GRAHAM, B. S., G. W. IMBENS, AND G. RIDDER (2014): "Complementarity and Aggregate Implications of Assortative Matching: A Nonparametric Analysis," *Quantitative Economics*, 5, 29–66. [719]

GRUETTER, M., AND R. LALIVE (2009): "The Importance of Firms in Wage Determination," *Labour Econ.*, 16, 149–160. [699]

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

HAGEDORN, M., T. H. LAW, AND I. MANOVSKII (2017): "Identifying Equilibrium Models of Labor Market Sorting," *Econometrica*, 85, 29–65. [700,702,705,712,722]

HAHN, J., AND H. R. MOON (2010): "Panel Data Models With Finite Number of Multiple Equilibria," *Econometric Theory*, 26, 863–881. [702,733]

HALL, P., AND X.-H. ZHOU (2003): "Nonparametric Estimation of Component Distributions in a Multivariate Mixture," *Ann. Stat.*, 31, 201–224. [702]

HALL, R. E., AND F. S. MISHKIN (1982): "The Sensitivity of Consumption to Transitory Income: Estimates From Panel Data on Households," *Econometrica*, 50, 461–481. [706]

HENRY, M., Y. KITAMURA, AND B. SALANIÉ (2014): "Partial Identification of Finite Mixtures in Econometric Models," *Quant. Econom.*, 5, 123–144. [702]

HU, Y. (2008): "Identification and Estimation of Nonlinear Models With Misclassification Error Using Instrumental Variables: A General Solution," *Journal of Econometrics*, 144, 27–61. [702]

HU, Y., AND S. M. SCHENNACH (2008): "Instrumental Variable Treatment of Nonclassical Measurement Error Models," *Econometrica*, 76, 195–216. [702]

HU, Y., AND M. SHUM (2012): "Nonparametric Identification of Dynamic Models With Unobserved State Variables," *J. Econom.*, 171, 32–44. [702]

HWANG, H.-S., D. T. MORTENSEN, AND W. R. REED (1998): "Hedonic Wages and Labor Market Search," *Journal of Labor Economics*, 16, 815–847. [720]

JACKSON, C. K. (2013): "Match Quality, Worker Productivity, and Worker Mobility: Direct Evidence From Teachers," *Rev. Econ. Stat.*, 95, 1096–1116. [700]

JOCHMANS, K., AND M. WEIDNER (2017): "Fixed-Effect Regressions on Network Data," Unpublished Manuscript. [717]

KLINE, P., R. SAGGIO, AND M. SØLVSTEN (2018): "Leave-out Estimation of Variance Components," ArXiv e-prints. [701]

KRAMARZ, F., S. J. MACHIN, AND A. OUAZAD (2015): "Using Compulsory Mobility to Identify School Quality and Peer Effects," *Oxford Bulletin of Economics and Statistics*, 77, 566–587. [700]

LAMADON, T., J. LISE, C. MEGHIR, AND J. M. ROBIN (2013): "Matching, Sorting, Firm Productivity and Wages," Unpublished Manuscript. [700]

LENTZ, R., S. PIYAPROMDEE, AND J.-M. ROBIN (2017): "On Worker and Firm Heterogeneity in Wages and Employment Mobility: Evidence From Danish Register Data," Unpublished Manuscript. [728]

LEVINE, M., D. R. HUNTER, AND D. CHAUVEAU (2011): "Maximum Smoothed Likelihood for Multivariate Mixtures," *Biometrika*, 98, 403–416. [702]

LIN, C.-C., AND S. NG (2012): "Estimation of Panel Data Models With Parameter Heterogeneity When Group Membership Is Unknown," *Journal of Econometric Methods*, 1. [702]

LINDENLAUB, I. (2017): "Sorting Multidimensional Types: Theory and Application," *The Review of Economic Studies*, 84, 718–789. [720]

LISE, J., AND F. POSTEL-VINAY (2015): "Multidimensional Skills, Sorting, and Human Capital Accumulation," Manuscript, University of Minnesota. [720]

LISE, J., AND J.-M. ROBIN (2013): "The Macro-Dynamics of Sorting Between Workers and Firms," *American Economic Review*. To appear. [706]

LISE, J., C. MEGHIR, AND J.-M. ROBIN (2016): "Matching, Sorting and Wages," *Review of Economic Dynamics*, 19, 63–87. [700,726]

LOPES DE MELO, R. (2018): "Firm Wage Differentials and Labor Market Sorting: Reconciling Theory and Evidence," *Journal of Political Economy*, 126, 313–346. [700]

MENDES, R., G. J. VAN DEN BERG, AND M. LINDEBOOM (2010): "An Empirical Assessment of Assortative Matching in the Labor Market," *Labour Econ.*, 17, 919–929. [699]

POSTEL-VINAY, F., AND J.-M. ROBIN (2002): "Equilibrium Wage Dispersion With Worker and Employer Heterogeneity," *Econometrica*, 70, 2295–2350. [702,705,706,712,726]

REIERSØL, O. (1950): "Identifiability of a Linear Relation Between Variables Which Are Subject to Error," *Econometrica*, 18, 375–389. [707]

SHI, S. (2008): "Directed Search for Equilibrium Wage-Tenure Contracts," *Econometrica*, 77, 561–584. [705]

SHIMER, R. (2005): "The Assignment of Workers to Jobs in an Economy With Coordination Frictions," *Journal of Political Economy*, 113, 996–1025. [705]

SHIMER, R., AND L. SMITH (2000): "Assortative Matching and Search," *Econometrica*, 68, 343–369. [700,701, 705,707,709,715,720]

SKANS, O. N., P.-A. EDIN, AND B. HOLMLUND (2009): "Wage Dispersion Between and Within Plants: Sweden 1985–2000," in *The Structure of Wages: An International Comparison*, ed. by E. P. Lazear and K. L. Shaw. University of Chicago Press, 217–260. [735,736]

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